

INTERNSHIP TASK REPORT

Cybersecurity & Ethical Hacking Internship Program

ApexPlanet Software Pvt. Ltd.

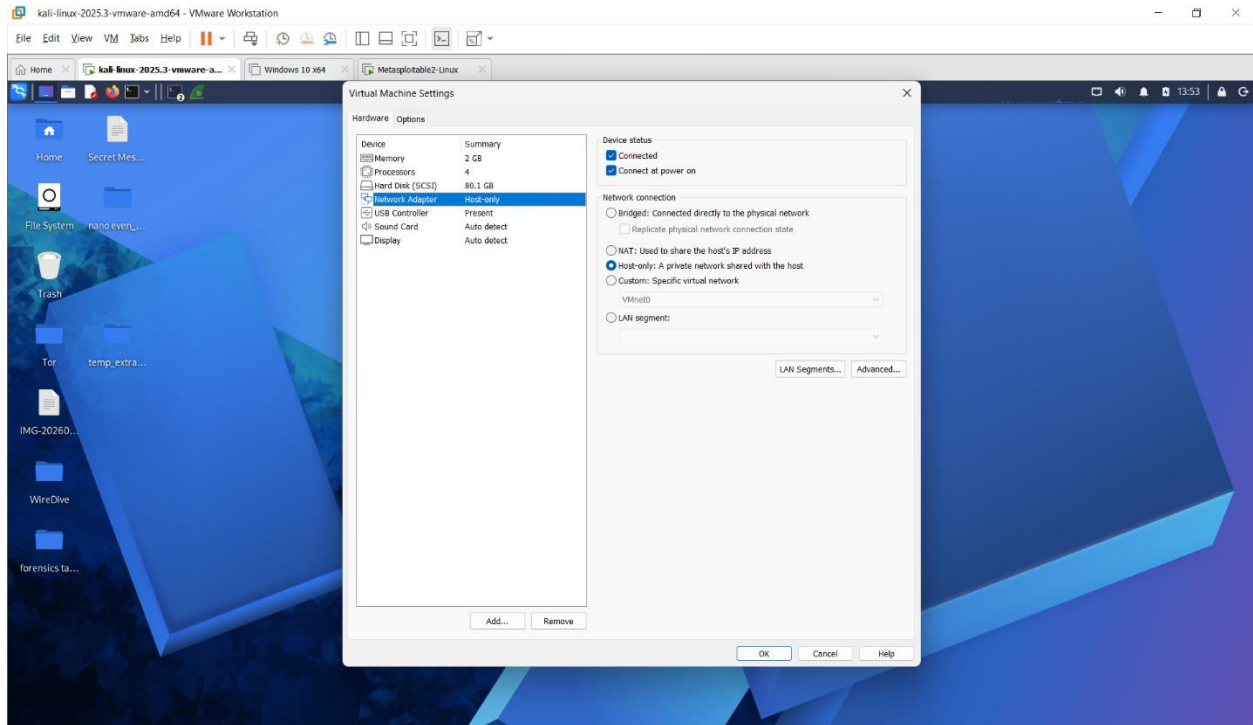
Submitted by :

- Student Name : Abhik Mondal
- Email ID : mondalabhik13@gmail.com
- Offer Letter ID : APSPL2646918
- Submission Date : 25/09/2026

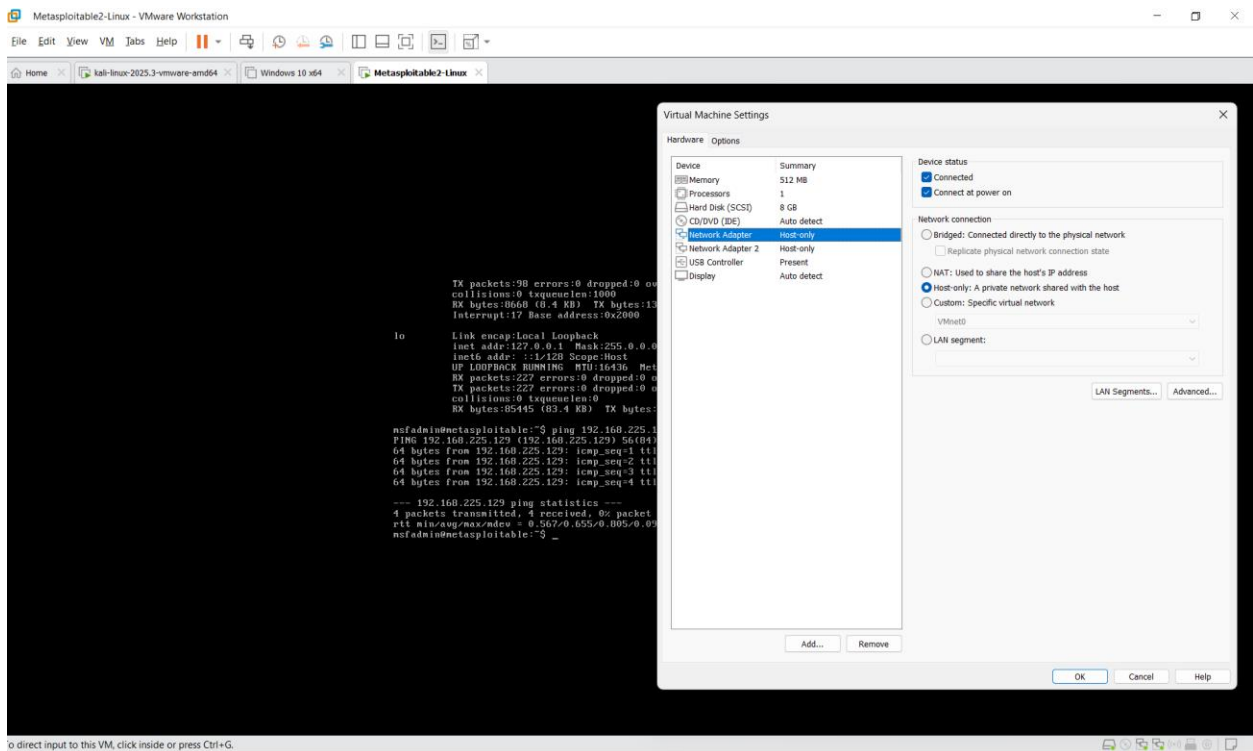
Executive Summary / Overview:

This report documents the completion of Task 1 for the Cybersecurity Internship. It covers the setup of a virtual penetration testing lab environment consisting of Kali Linux (Attacker Machine) and Metasploitable 2 (Target Machine) on VMware, configured within a Host-Only private network, along with initial packet capture verification using Wireshark.

Kali (Attacker Machine network adapter connect with Host only)



Metasploitable2 (Victim machine network adapter connect with Host only)



Ip of the victim machine 192.168.225.128

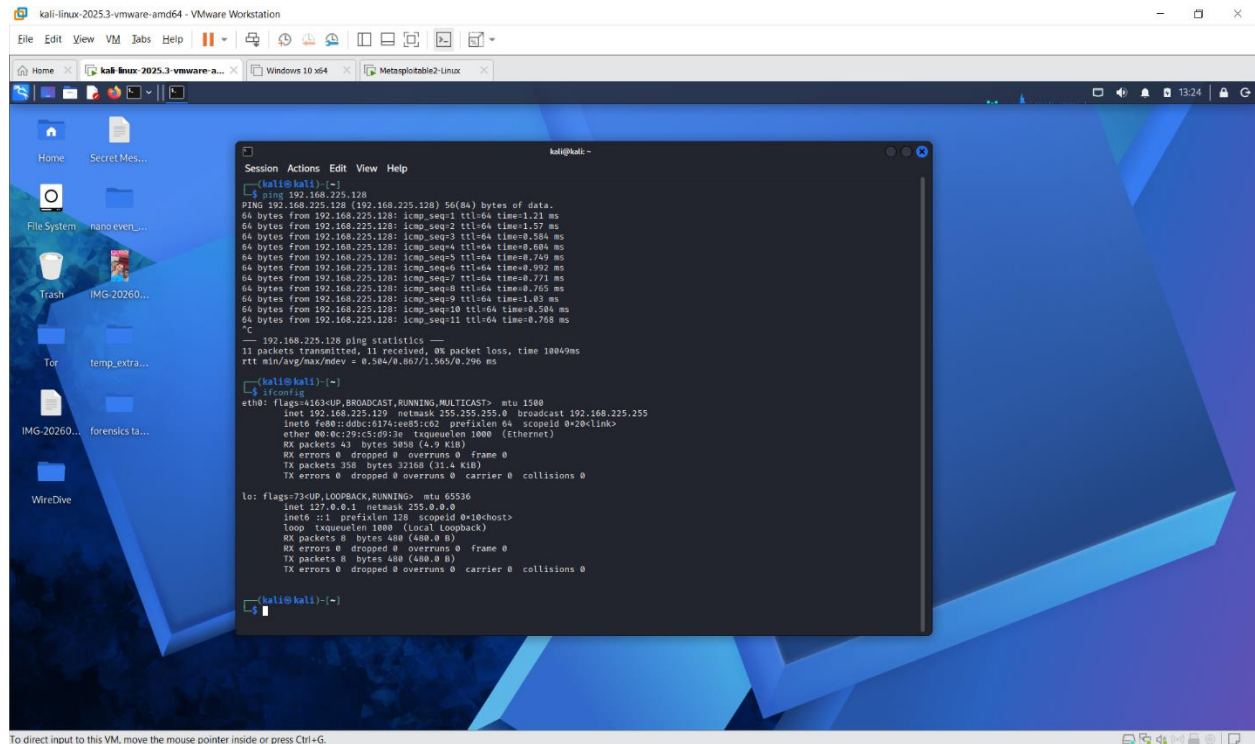
```
Metasploitable2-Linux - VMware Workstation
File Edit View VM Tabs Help
Home x kali-linux-2023.3-vmware-amd64 x Windows 10 x64 x Metasploitable2-Linux x

--- 192.168.225.129 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2000ms
rtt min/avg/max/mdev = 0.446/0.465/0.480/0.022 ms
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:33:69:b9
          inet addr:192.168.225.128  Bcast:192.168.225.255  Mask:255.255.0
          inet6 addr: fe80::20c:29ff:fe33:69b9/64  Scope:link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:150  errors:0  dropped:0  overruns:0  frame:0
          TX packets:190  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:8668 (8.4 KB)  TX bytes:13209 (12.9 KB)
          Interrupt:17  Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:222  errors:0  dropped:0  overruns:0  frame:0
          TX packets:222  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:0
          RX bytes:85445 (83.4 KB)  TX bytes:85445 (83.4 KB)

msfadmin@metasploitable:~$ _
```

Ip of the attacker machine 192.168.225.129
And also ping from attacker to victim machine



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following commands and output:

```
kali@kali:~$ ping 192.168.225.128
PING 192.168.225.128 (192.168.225.128) 56(84) bytes of data:
64 bytes from 192.168.225.128: icmp_seq=1 ttl=64 time=1.121 ms
64 bytes from 192.168.225.128: icmp_seq=2 ttl=64 time=1.57 ms
64 bytes from 192.168.225.128: icmp_seq=3 ttl=64 time=0.584 ms
64 bytes from 192.168.225.128: icmp_seq=4 ttl=64 time=0.584 ms
64 bytes from 192.168.225.128: icmp_seq=5 ttl=64 time=0.749 ms
64 bytes from 192.168.225.128: icmp_seq=6 ttl=64 time=0.592 ms
64 bytes from 192.168.225.128: icmp_seq=7 ttl=64 time=0.771 ms
64 bytes from 192.168.225.128: icmp_seq=8 ttl=64 time=0.765 ms
64 bytes from 192.168.225.128: icmp_seq=9 ttl=64 time=1.63 ms
64 bytes from 192.168.225.128: icmp_seq=10 ttl=64 time=0.584 ms
64 bytes from 192.168.225.128: icmp_seq=11 ttl=64 time=0.768 ms
^C
--- 192.168.225.128 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10049ms
rtt min/avg/max/mdev = 0.364/0.807/1.585/0.290 ms

kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.225.129 netmask 255.255.255.0 broadcast 192.168.225.255
    inet6 fe80::d6dc:c174:ee55:c02 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:c5:d9:3e txqueuelen 1000 (Ethernet)
    RX packets 43 bytes 3856 (4.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 358 bytes 32168 (31.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<lo>host>
    loop txqueuelen 1000 (Local loopback)
    RX packets 0 bytes 400 (400.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 400 (400.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl-G.

Ping from victim to attacker

```
Metasploitable2-Linux - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware-ami64 Windows 10 x64 Metasploitable2-Linux

TX packets:98 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:8668 (8.4 KB) TX bytes:13289 (12.9 KB)
Interrupt:17 Base address:0x2000

lo Link encap:Local Loopback
  inet addr:127.0.0.1 Bcast:255.0.0.0
  inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:65536 Metric:1
RX packets:227 errors:0 dropped:0 overruns:0 frame:0
TX packets:227 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:85445 (83.4 KB) TX bytes:85445 (83.4 KB)

msfadmin@metasploitable2:~$ ping 192.168.225.129
PING 192.168.225.129 (192.168.225.129) 56(84) bytes of data.
64 bytes from 192.168.225.129: icmp_seq=1 ttl=64 time=0.567 ms
64 bytes from 192.168.225.129: icmp_seq=2 ttl=64 time=0.453 ms
64 bytes from 192.168.225.129: icmp_seq=3 ttl=64 time=0.895 ms
64 bytes from 192.168.225.129: icmp_seq=4 ttl=64 time=0.536 ms

--- 192.168.225.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2997ms
rtt min/avg/max/mdev = 0.567/0.655/0.895/0.693 ms
msfadmin@metasploitable2:~$ _
```

Capture of network packet through check wireshark

The image shows a Wireshark network packet capture. The top pane displays a list of 21 ICMP packets. The middle pane shows the details of the selected packet (No. 21), including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
56	21.472196165	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
57	22.496419828	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
58	22.496946587	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
59	23.528437752	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
60	23.521028269	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
61	24.544140888	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
62	24.544822456	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
63	25.567043637	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
64	25.568589656	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
67	26.591682774	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
68	26.592119455	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
69	27.616168119	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request
70	27.616811524	192.168.225.128	192.168.225.129	ICMP	98	Echo (ping) reply
71	28.648375337	192.168.225.129	192.168.225.128	ICMP	98	Echo (ping) request

Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
Ethernet II, Src: VMware_c5:d9:3e (00:0c:29:c5:d9:3e), Dst: VMware_33:69:b9 (00:0c:29:33:69:b9)
Internet Protocol Version 4, Src: 192.168.225.129, Dst: 192.168.225.128
Internet Control Message Protocol

Packets: 71 - Displayed: 57 (80.3%)

#List all files and directories with details, #And also change file permissions

```
(root@kali):~# ls -la
total 184
drwx-- 25 root root 4096 Sep 23 13:39 .
drwxr-xr-x 18 root root 4096 Aug 19 09:01 ..
drwxr-xr-x 5 root root 4096 Aug 8 07:52 asm-ai-engine
-rw-r--r-- 1 root root 17 Mar 29 04:26 .bash_history
-rw-r--r-- 1 root root 228 Nov 7 2025 .bash_logout
-rw-r--r-- 1 root root 5551 Sep 9 2025 .bashrc
-rw-r--r-- 1 root root 687 Sep 9 2025 .bashrc.original
drwx-- 16 root root 4096 Sep 23 13:39 .cache
drwxrwxr-x 6 root root 4096 Nov 17 2025 Cavellion
drwxrwxr-x 17 root root 4096 Sep 21 08:10 .config
drwx-- 3 root root 4096 Nov 2 2025 .dbus
drwxr-xr-x 2 root root 4096 Nov 7 2025 Desktop
-rw-r--r-- 1 root root 35 Nov 7 2025 .dmrc
drwxr-xr-x 2 root root 4096 Nov 7 2025 Documents
drwxr-xr-x 2 root root 4096 Mar 29 18:18 Downloads
-rw-r--r-- 1 root root 11656 Sep 9 2025 .face
lrwxrwxrwx 1 root root 11 Aug 19 09:11 .face.icon -> /root/.face
drwx-- 3 root root 4096 Aug 7 2025 .fnm
drwx-- 2 root root 4096 Nov 2 2025 .gifs
-rw-r--r-- 1 root root 0 Nov 7 2025 .ICAuthority
drwxr-xr-x 3 root root 4096 Nov 7 2025 .java
drwx-- 5 root root 4096 Nov 7 2025 .local
drwx-- 4 root root 4096 Nov 7 2025 .mozilla
drwxr-xr-x 2 root root 4096 Nov 7 2025 Music
drwxrwxr-x 6 root root 4096 Jan 9 2026 PhoneInfo
drwxr-xr-x 2 root root 4096 Nov 7 2025 Pictures
drwxr-xr-x 2 root root 4096 Apr 1 14:42 practice-lab
-rw-r--r-- 1 root root 132 Aug 17 2025 .profile
drwxr-xr-x 2 root root 4096 Sep 21 08:10 Projects
drwxr-xr-x 2 root root 4096 Nov 7 2025 Public
drwxr-xr-x 2 root root 4096 Mar 27 05:11 .ssh
drwxr-xr-x 2 root root 4096 Nov 7 2025 Templates
drwxr-xr-x 2 root root 4096 Nov 7 2025 Videos
-rw-r--r-- 1 root root 98 Sep 21 08:10 .Authority
-rw-r--r-- 1 root root 4351 Sep 21 08:10 .xsession-errors
-rw-r--r-- 1 root root 4345 Aug 8 07:51 .xsession-errors.old
drwxrwxr-x 8 root root 4096 Nov 13 2025 .zipster
-rw-r--r-- 1 root root 336 Nov 7 2025 .zprofile
-rw-r--r-- 1 root root 4438 Aug 8 08:01 .zsh_history
-rw-r--r-- 1 root root 18855 Sep 9 2025 .zshrc

(root@kali):~# chmod 777 asm-ai-engine
(root@kali):~#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Show active listening ports

```
(root@kali):~# chmod 777 asm-ai-engine
(root@kali):~#
(root@kali):~#
(root@kali):~#
(root@kali):~#
(root@kali):~# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp6       0      0 :::22                  :::*                    LISTEN

(root@kali):~# ping -c 4 192.168.225.128
PING 192.168.225.128 (192.168.225.128) 56(84) bytes of data:
64 bytes from 192.168.225.128: icmp_seq=1 ttl=64 time=0.76 ms
64 bytes from 192.168.225.128: icmp_seq=2 ttl=64 time=0.683 ms
64 bytes from 192.168.225.128: icmp_seq=3 ttl=64 time=0.745 ms
64 bytes from 192.168.225.128: icmp_seq=4 ttl=64 time=0.711 ms

--- 192.168.225.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 384ms
rtt min/avg/max/mdev = 0.682/2.954/9.768/3.929 ms
(root@kali):~#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Check connectivity to a target ip, and also check service version -sV

```
root@kali:~#
Session Actions Edit View Help
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                     LISTEN

root@kali:~# ping -c 4 192.168.225.128
PING 192.168.225.128 (192.168.225.128) 56(84) bytes of data:
64 bytes from 192.168.225.128: icmp_seq=1 ttl=64 time=0.76 ms
64 bytes from 192.168.225.128: icmp_seq=2 ttl=64 time=0.602 ms
64 bytes from 192.168.225.128: icmp_seq=3 ttl=64 time=0.745 ms
64 bytes from 192.168.225.128: icmp_seq=4 ttl=64 time=0.711 ms
--- 192.168.225.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3043ms
rtt min/avg/max/mdev = 0.602/0.754/0.760/3.929 ms

root@kali:~# nmap -sV 192.168.225.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-23 13:45 -0400
Nmap scan report for 192.168.225.128
Host is up (0.802s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rsh           rsh (RFC 4100)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?       netkit-rsh
514/tcp   open  shell        netkit-rsh
1009/tcp  open  java-rmi     GNU Classpath gmrregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1.1
3180/tcp  open  mysql        MySQL 5.0.51a-ubuntu5
5632/tcp  open  postgresql   PostgreSQL 9B 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6080/tcp  open  X11          (access denied)
6087/tcp  open  irc          UnrealIRCd
8080/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC address: 08:00:27:13:16:8B (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: o:/linux/linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 57.43 seconds

root@kali:~#
```

AES-256 Encryption & Decryption

```
kali@kali:~#
Session Actions Edit View Help

kali@kali:~# echo "Hello Cybersecurity" > secret.txt

kali@kali:~# ls
bapi_xc.txt  bruteforce  CatVision  Desktop  Downloads  encrypt.py  Lz5Egg8b.html  maxphisher-updated  Mr.Holmes  phoneinfoga  Pictures  Public  spiderfoot  Templates  venv  zphisher
bindhack.exe  Bruteforce  Desktop  Downloads  journeywithabhi.txt  MaxPhisher  Media  Music  PhoneInfoga  Projects  secret.txt  sweet.exe  The-Black-Tiger  Videos

kali@kali:~# openssl enc -aes-256-cbc -pbkdf2 -salt -in secret.txt -out secret.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:

kali@kali:~# openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out decrypted.txt
enter AES-256-CBC decryption password:

kali@kali:~# ls
bapi_xc.txt  bruteforce  CatVision  Desktop  Downloads  journeywithabhi.txt  MaxPhisher  Media  Music  PhoneInfoga  Projects  secret.enc  spiderfoot  Templates  venv  zphisher
bindhack.exe  Bruteforce  Desktop  Downloads  journeywithabhi.txt  MaxPhisher  Media  Music  PhoneInfoga  Projects  secret.txt  sweet.exe  The-Black-Tiger  Videos

kali@kali:~# cat decrypted.txt
Hello Cybersecurity

kali@kali:~#
```

Update system repositories

```
kali@kali: ~  
$ sudo apt update  
[sudo] password for kali:  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Err:1 http://http.kali.org/kali kali-rolling InRelease  
Temporary failure resolving 'http.kali.org'  
96 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Temporary failure resolving 'http.kali.org'  
Warning: Some index files failed to download. They have been ignored, or old ones used instead.  
Upgrading:  
at-spi2-common libasan8 libexpat1-dev libldap-common libqt5network6 libqt5xml6 librtld1 python3-trio systemd-tpm xserver-xorg-legacy  
at-spi2-core libatk-bridge2.0-0t64 libgdm1 libldsp2 libqt5opengl6 libquda0 librtld1-common python3-typing-inspection systemd-userdbd xzfb  
glibc-4 libatk1-0t64 libgcc-s1 liblsan0 libqt5openglwidgets6 libstdc++6 libzstd1 python3-uhjson udev xad xxd zsh  
gcc-10-base libatomic1 libgfortran5 libmodel37 libqt5printsupport6 libsystemd-shared mesa-libgallium qt5-base-dev-tools vim vim-common zsh-common  
girl2-attk-1.0 libatspi2.0-0t64 libgl1-mesa-dri libmss-systemd libqt5sol6 libsystemd0 mesa-vulkan-drivers qt5-gtk-platformtheme vim-runtime zsh-common  
girl2-attk-2.0 libbytesize-common libgl-mesa0 libobjjs libqt5sql-sqlite libxcan2 nodejs qt5-qpa-plugins vim-tiny vim-tiny-common xserver-common  
girl2-vte-2.91 libbytesize1 libgomp1 libpaan-systemd libqt5test6 libubian1 nodejs-doc ruby-json vim-tiny vim-tiny-common xserver-common  
ldap-urllib libcc1-0 libmcsan0 libqt5core6t64 libqt5waylandclients libudev1 python3-filelock systemd python3-humansize systemd-timesyncd xserver-xorg-core  
libqt5gui4 libgcc-mesa0 libitm1 libqt5dbus6 libqt5widgets6 libvte-2.91-0 python3-python-socks systemd-sysv xserver-xorg-core  
libqt5xdp6 libexpat1 libjs-sphinxdoc libqt5gui6 libqt5shellintegration6 libvte-2.91-common python3-python-socks systemd-timesyncd xserver-xorg-core  
Summary:  
Upgrading: 96, Installing: 0, Removing: 0, Not Upgrading: 0  
Download size: 127 MB  
Space needed: 15.7 MB / 47.7 GB available  
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 libsystemd0 amd64 261.2-1  
Ign:2 http://http.kali.org/kali kali-rolling/main amd64 xserver-common all 2:21.1.24-1  
OK [Connecting to http.kali.org]
```

SHA-256 Hashing of the secret.txt

```
kali@kali: ~  
$ sha256sum secret.txt  
b0ba7d7284c3373619c3c8a3d98d761eab3f0efc76484ef1f9b9a4045a9e4a8e secret.txt  
$
```